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END TERM PRACTICAL EXAMINATIONS – Nov-Dec 2025

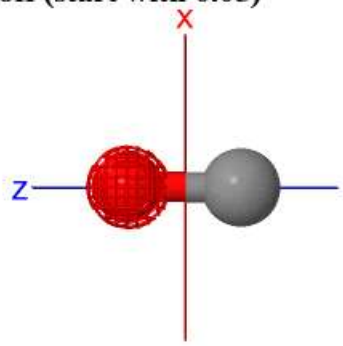
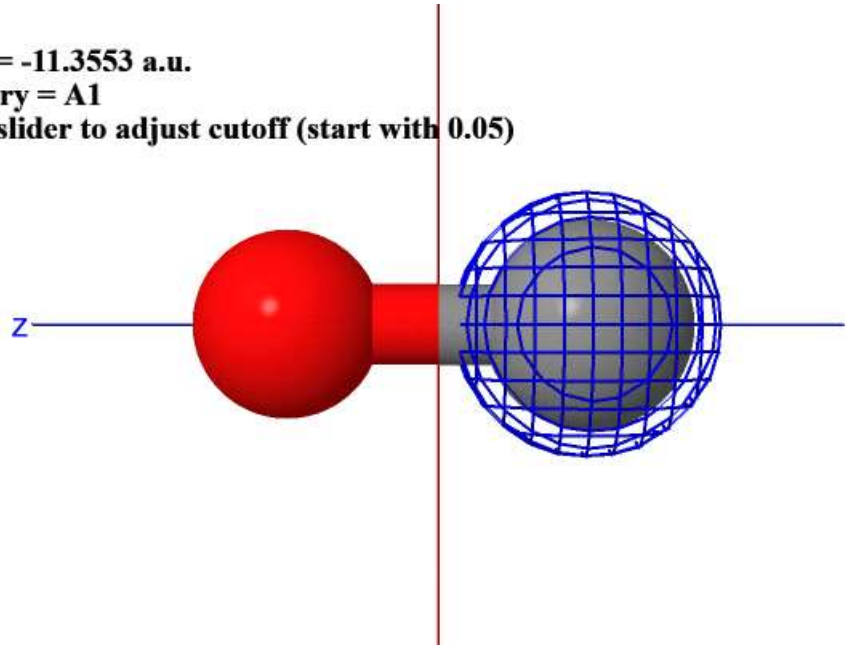
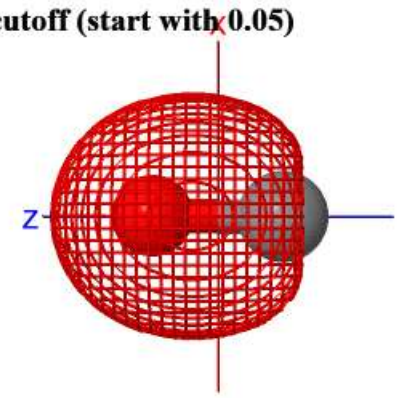
Programme	:	**B.Tech.**	Semester		**Fall Seme**
Course Title/Course Code	:	**Introduction to Computational Chemistry/CHY1005**	Slot		
Time	:	**1 ½ hours**	Max. Marks	:	**30**

Answer all the Questions

Q.No.	Sub. Sec.	Question Description

1		Draw the molecules and calculate the optimized energy for the following molecules:

Molecule	Optimized Energy	Bond Length
NH_3	-56.18436 a.u.	0.102
H_2O	-76.01075 a.u	0.097
CF_4	-435.64521 a.u.	0.133
HI	-6918.43935 a.u.	0.164

2		Draw the molecular orbital diagram for the following molecules: • CO **MO 1/17** **Energy = -20.675 a.u.** **Symmetry = A1** **Use the slider to adjust cutoff (start with 0.05)**  • **MO 2/17** **Energy = -11.3553 a.u.** **Symmetry = A1** **Use the slider to adjust cutoff (start with 0.05)**  • **MO 3/17** **Energy = -1.5292 a.u.** **Symmetry = A1** **Use the slider to adjust cutoff (start with 0.05)** 

3	(a)	Calculate the Enthalpy of formation for the following reactions:

- $H_2 + F_2 = 2HF$

$\Delta H_f\ (H_2) = -1.15$ kCAL/MOL

$\Delta H_f\ (F_2) = 1.05$ KCAL/MOL

$\Delta H_f\ (HF) = -65.53$ kCAL/MOL

$2HF = -131.06$ KCAL/MOL

$\Delta H_f = \Delta H_f$ (Product) $- \Delta H_f$ (Reactant)

$\Delta H_f = -131.06$ KCAL/MOL $- (-1.15$ KCAL/MOL $+ 1.05)$

$= -130.96$ KCAL/MOL

- $N_2 + 3H_2 \rightleftharpoons 2NH_3$
-
- $C_4H_{10} + 6.5O_2 = 4CO_2 + 5H_2O$
-
- $N_2 + 3H_2 = 2NH_3$

-130.96 KCAL/MOL

3) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

$\Delta H\ N_2 = 2.11$ KCAL/MOL

$\Delta H\ H_2 = -1.15$ kCAL/MOL $\times 3 = -3.45$ kCAL/MOL

$\Delta H\ NH_3 = -10.08$ kCAL/MOL $\times 2$

$= -20.16$ kCAL/MOL

$\Delta H_{total} = \Delta H$ product $- \Delta H$ reactant

$= -20.16 - (2.11 + -3.45)$

$\Delta H_{total} = -18.82$ KCAL/MOL

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-
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4		Calculate the IR stretching frequency for the following molecules: • C_2H_2 = (882,200) • and b) CH_3CHO = 2.03e + 3,222